

COMPLETE DOCUMENTATION

Manufacturing Module

Raw Materials · Bill of Materials · Work Orders · Product Batches · Costing

CRM Module Documentation — Three Audiences

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For Sellers

How to pitch, position, and sell the Manufacturing module

1.1 The 30-Second Pitch

Most CRMs stop at "we sold something." This CRM's Manufacturing module tracks what it takes to **make** that something — raw materials, production runs, real production cost, and profit margin — all inside the same CRM the customer already uses for leads and invoices.

ONE-LINER FOR THE PROSPECT

"If you buy raw material, make a product from it, and sell it — this module tracks the whole chain and tells you your real profit, not just your sale price."

1.2 Who To Pitch This To

GOOD FIT

Small/mid manufacturers, workshops, furniture makers, food processors, garment units, fabricators — anyone who converts raw material into a sellable product in-house.

NOT A FIT (YET)

Pure trading/reseller businesses (nothing to "make"), businesses needing multi-stage sub-assembly production, or businesses needing shop-floor/labor scheduling — see Part 3 roadmap.

1.3 Pain Points This Solves

- 1 "I don't actually know if I'm making money on this product." Most small manufacturers price by gut feeling. This module computes real production cost (material + labor + machine) and shows margin against the selling price automatically.

- 2 **"We keep running out of raw material without warning."** Low Stock alerts fire automatically, and a one-click button turns the shortfall into a ready-to-send Purchase Request — no manual math.
- 3 **"We have no idea what's expiring in the warehouse."** Product Batches track expiry per lot and alert before stock goes bad — relevant for food, chemicals, cosmetics, pharma-adjacent goods.
- 4 **"Excel/WhatsApp is how we track production right now."** Work Orders give a simple, auditable Pending → In Progress → Completed flow that any staff member can run without training.

1.4 What's Included

FEATURE	WHAT THE CUSTOMER GETS
Raw Material vs Finished Good	Every product is tagged so the system knows what's an input vs. a sellable output.
Bill of Materials (BOM)	Define exactly what raw materials (and how much of each) go into one unit of a finished product.
Work Orders	Log a production run: pick the product, quantity, hit Start then Complete — stock moves automatically, blocked if materials are short.
Production Costing	Material cost auto-calculated and frozen at completion; labor and machine cost entered manually; total cost, cost-per-unit, and margin all computed.
Product Batches	Every batch of stock (from a purchase or a production run) is tracked with expiry date and consumed oldest-expiry-first automatically.
Low Stock → Purchase Request	One click turns a material shortage into a ready-to-submit purchase request, pre-filled with the right quantities.

1.5 Packaging & How To Grant Access

Manufacturing is a **Pro-plan feature**. Two ways a customer gets it:

- 1 **They're on (or upgrade to) the Pro plan** — access turns on automatically, no manual step needed.
- 2 **Superadmin grants it manually as an exception** — for a demo, a trial, or a customer on a lower plan who needs to evaluate it before committing to an upgrade. Superadmin → Tenants → open the tenant → "Module Access" card → Enable.

SALES TIP

Use the manual-grant path to give a live demo on a prospect's own account before asking them to upgrade

— it removes "will it actually work for us" as an objection.

1.6 Objection Handling

OBJECTION	RESPONSE
"We already use Excel for this."	Excel doesn't stop you from completing a production run when you don't have enough material, doesn't auto-calculate margin, and doesn't alert you before stock expires. This does all three, in the same place as your sales data.
"We have complex multi-stage production."	Be upfront: today's version supports single-level BOM (raw materials → one finished good). Multi-stage sub-assembly is on the roadmap (see Part 3) — good to flag as "coming soon," not oversell it as already there.
"Can we track which worker/machine did the job?"	Not yet — shop-floor/labor tracking is roadmap, not built. Be honest; don't oversell.



For End Users

A practical guide for Tenant Admins and Staff

2.1 Before You Start — Is Manufacturing Turned On?

Manufacturing is not visible by default. If you don't see "**Work Orders**" in the left sidebar under the Purchase section, it isn't enabled for your account yet — it comes with the Pro plan, or your CRM provider can turn it on for you as an exception.

2.2 Step 1 — Set Up Your Raw Materials

- 1 Go to **Products** → **Add Product**.
- 2 Fill in Name, Rate (this is your purchase cost for a raw material), Unit (Kg, Litre, Metre, etc.).
- 3 Set **Type** = **Raw Material**.
- 4 Set **Current Stock** (what you have on hand right now) and, optionally, a **Reorder Level** + **Reorder Quantity** so the system can alert you before you run out.

2.3 Step 2 — Set Up Your Finished Goods

Same Add Product screen, but set **Type** = **Finished Good**. This is what you sell to customers on Invoices/Quotations.

2.4 Step 3 — Define the Recipe (Bill of Materials)

Open a Finished Good product's **Edit** page — a "Bill of Materials" section appears. Add a row per raw material it needs, and how much of that material is needed to make **one unit** of the finished good.

EXAMPLE

Product: "Steel Chair". BOM: 5 Kg of "Steel Rod" per chair. So a Work Order for 10 chairs needs 50 Kg of Steel Rod.

2.5 Two Ways Your Stock Moves

METHOD	WHEN TO USE IT
Automatic (Invoice sale)	You don't run formal production batches — you just make-to-order. The moment you invoice a Finished Good, its BOM raw materials are deducted automatically, no extra step. This works even without Manufacturing turned on.
Work Orders	You run actual production batches ahead of selling (e.g. "make 50 chairs this week"). Gives you a formal Pending → In Progress → Completed record, blocks completion if material is short, and lets you log labor/machine cost. Requires Manufacturing to be enabled.

2.6 Running a Work Order

- 1 Create** — Work Orders → New. Pick the Finished Good and the quantity to produce. The screen previews exactly which raw materials will be needed.
- 2 Start Production** — moves it to "In Progress." This just marks that work has begun; stock is untouched at this point.
- 3 Mark Completed** — this is the real moment: raw materials are deducted (oldest-batch-first if you're tracking batches), and the finished good's stock goes up by the quantity produced. If you don't have enough raw material, the system blocks completion and tells you exactly what's short and by how much.
- 4 Cancel** — available any time before completion, no stock effect.

CAREFUL

Once a Work Order is "In Progress," you can no longer change the product or quantity — only Start/Complete/Cancel it. Get the quantity right before starting.

2.7 Understanding Production Cost & Margin

On a Work Order's detail page, once it's completed, you'll see:

- **Material Cost** — auto-calculated from the BOM at the moment of completion, then **frozen** — it won't change even if raw material prices go up later.

- **Labor Cost / Machine Cost** — enter these yourself once you know the actual figures (you can update them any time, even after completion).
- **Total Cost** and **Cost per Unit** — added up automatically.
- **Margin** — compares your total cost against the product's current selling rate, so you instantly see if you're actually making money.

2.8 Low Stock Alerts & Purchase Requests

- 1 When any product (raw material or finished good) drops to or below its Reorder Level, it appears on the **Low Stock** page (under Products), and admins get an in-app notification.
- 2 For a Finished Good, the page shows exactly which raw materials — and how much — you need to restock, calculated from its BOM.
- 3 Click "**Create Purchase Request**" — it opens a pre-filled request with the right materials and quantities. Just review and submit.

2.9 Product Batches & Expiry

Every time stock comes in (a Purchase Order is received, or a Work Order is completed), it's recorded as a "batch" with its own received date and optional expiry date. When stock goes out (a sale, or raw material consumed by a Work Order), the system automatically uses the **batch closest to expiring first** (FEFO — First-Expiry-First-Out). View a product's batches from the Products list ("Batches" button) — batches nearing expiry get flagged automatically.

2.10 Frequently Asked Questions

QUESTION	ANSWER
Can I edit the recipe (BOM) after I've already produced something?	Yes — but it only affects future Work Orders. Already-completed ones keep their frozen cost.
What if I don't set an expiry date on a batch?	It's treated as never-expiring and consumed last, after all dated batches.
Can staff without "Manage" permission run Work Orders?	They can create one if they have Create permission, but Start/Complete/Cancel needs the Manage permission — ask your admin.



For Developers

Architecture, data model, and roadmap

3.1 Architecture Overview

The Manufacturing module extends the existing `Product` model rather than introducing a parallel catalog. A product is either a `finished_good` (sellable) or a `raw_material` (consumed), controlled by `Product.type`. Everything else — BOM, Work Orders, Batches, Stock movement — hangs off that single distinction.

LAYER	FILES
Models	<code>Product</code> (extended), <code>BillOfMaterialItem</code> , <code>WorkOrder</code> , <code>ProductBatch</code> , <code>Tenant</code> (module gating)
Services	<code>StockService</code> (all stock movement — sales, receiving, BOM consumption, batch FEFO), <code>WorkOrderService</code> (production lifecycle)
Controllers	<code>ProductController</code> (BOM CRUD embedded in product edit, Low Stock, Batches), <code>WorkOrderController</code>
Access control	<code>EnsureModuleEnabled</code> middleware (<code>module:manufacturing</code>), <code>WorkOrderPolicy</code>

3.2 Data Model

PRODUCT (EXTENDED COLUMNS)

```
type enum(finished_good, raw_material) default finished_good
current_stock decimal(12,2) default 0
reorder_level decimal(12,2) nullable
reorder_quantity decimal(12,2) nullable
```

BILL_OF_MATERIAL_ITEMS

```
tenant_id, product_id (finished good), material_id (raw material), quantity_per_unit decimal(12,4)
unique(product_id, material_id)
```

`Product::billOfMaterials()` — rows where the product is the finished good being built.

WORK_ORDERS

```
number (WO-YYYYMMDD-NNNN), product_id, quantity, status enum(pending, in_progress, completed, cancelled)
started_at, completed_at, notes, created_by
labor_cost, machine_cost decimal(12,2) default 0
material_cost_snapshot decimal(12,2) nullable – frozen at completion
```

PRODUCT_BATCHES

```
tenant_id, product_id, batch_number, quantity, initial_quantity
expiry_date, received_at, purchase_order_id, work_order_id, expiry_notified_at
```

Every stock increase (PO receive, Work Order completion, invoice-restore) creates a batch row via `StockService::receiveBatch()`. Every stock decrease consumes across batches oldest-expiry-first via `StockService::consumeBatchesFEFO()` — `Product.current_stock` is always the source of truth for "how much," batches are a traceability ledger on top.

3.3 StockService — Method Reference

METHOD	PURPOSE
<code>applyInvoiceItems(\$items, \$sign)</code>	Invoice create/update/delete hook. Negative sign = sale (FEFO consume + auto BOM consume); positive = restore.
<code>consumeBomForSale(\$product, \$qty)</code>	Auto-deducts a finished good's BOM raw materials on sale — the "no Work Order needed" path.
<code>suggestedMaterialsFor(\$product)</code>	BOM explosion for the Low Stock → Purchase Request flow.
<code>productionShortfallFor(\$product, \$qty)</code>	Pre-check used by <code>WorkOrderService::complete()</code> to block if material is short.
<code>consumeForProduction(\$product, \$qty, \$workOrderId, \$expiry)</code>	Consumes BOM materials FEFO and credits the finished good as a new batch, tagged to the Work Order.
<code>receiveBatch(...)</code> / <code>consumeBatchesFEFO(...)</code>	Low-level batch ledger primitives, shared by PO receiving, Work Orders, and invoice restores.

`lowStockProducts($tenantId)`

Products at/below reorder level, for the Low Stock page.

3.4 WorkOrderService — Lifecycle

`store()` → `start()` → `complete()` (or `cancel()` from either pending or in_progress). `complete()` : checks `productionShortfallFor()` , throws `ValidationException` if short; otherwise snapshots material cost, calls `consumeForProduction()` , marks completed.

3.5 Module Access Control

Tri-state per-tenant gating on `Tenant.hasModuleEnabled(string $module)` :

1. `settings['modules'][$module] === true` → forced ON (superadmin override)
2. `settings['modules'][$module] === false` → forced OFF (superadmin override)
3. key absent → falls back to `Plan.features[$module] === true`

Enforced via `module:manufacturing` route middleware (`EnsureModuleEnabled`) on the Work Orders route group and the Product Batches route. Superadmin UI: Tenants → tenant detail → "Module Access" card (Enable / Disable / Reset to Plan Default). Every toggle writes an `AuditLog` entry.

NOTE FOR FUTURE WORK IN THIS AREA

As of this writing, other in-progress work in the same codebase (Service Catalog, Subscriptions, Appointments, Time Tracking, Tickets) has begun generalizing this into a `TOGGLEABLE_MODULES` pattern in `TenantController` . The Manufacturing-specific toggle methods were deliberately left as-is rather than merged into that generic pattern, to avoid colliding with concurrent changes — reconciling them is a reasonable follow-up once that other work stabilizes.

3.6 Notifications

TYPE	FIRES WHEN
<code>product.low_stock</code>	Any product crosses at/below reorder level (invoice sale or BOM consumption)
<code>product.batch_expiring</code>	<code>SendBatchExpiryAlerts</code> console command, batches inside the alert window
<code>work_order.completed</code>	Work Order completes — notifies other tenant_admins / <code>work_orders.view_all</code> holders (not the completer)

3.7 Permissions

`work_orders.view_own` , `view_all` , `create` , `edit` , `manage` (Start/Complete/Cancel), `delete` . `edit` only while `pending` ; `manage` blocked once `completed` / `cancelled` . `editCosts` (labor/machine cost entry) uses `manage` permission but is **not** status-restricted, since real costs are usually known only after completion.

3.8 Known Limitations (as of this document)

- No multi-level BOM — a raw material cannot itself have a BOM (no sub-assemblies).
- No shop-floor tracking — no operator, machine/station, or time-log capture.
- No production capacity planning.
- Labor/Machine cost is flat manual entry, not hours × rate.
- No cost-variance reporting (estimate vs. actual across Work Orders).
- Manufacturing module gating is bundle-level (Work Orders + Batches together) — cannot grant one without the other yet.
- No usage-based/metered limits — access is binary on/off, not quota-based.

3.9 Roadmap — Not Yet Built

ITEM	WHAT IT WOULD DO	VALUE	EFFORT
Per-hour labor rate	Replace flat labor_cost with hours × rate/hour, reusable across Work Orders	Medium	Small
Cost variance report	Compare estimated vs. actual cost across Work Orders to spot systematic overruns	Medium	Medium
Granular module gating	Split the bundle into <code>manufacturing.work_orders</code> / <code>manufacturing.batches</code> so either can be granted independently — architecture is already tri-state and ready for this, just needs the split wired through routes + UI (see 3.5 note on coordinating with concurrent work first)	Low	Small
Multi-level BOM	A raw material can itself be built from other raw materials (sub-assemblies), with cycle detection	Situational	Large

Shop-floor tracking	Operator assignment, machine/station tracking, time logs per Work Order	Situational	Large
Production capacity planning	Check machine/resource availability before scheduling a Work Order for a given period	Situational	Large
Usage-based module limits	Meter access (e.g. "50 Work Orders/month") instead of a binary on/off	Low	Medium

This document covers the Manufacturing module (Bill of Materials, Work Orders, Product Batches, Production Costing, and Module Access Gating) as implemented through 22 August 2026. Part 3.5's note on concurrent work should be re-checked before extending the module gating system further.